

LU Decomposition

LU decomposition is the method of representing an $n \times n$ matrix A as $A = LU$, where L is lower triangular and U is upper triangular $\rightarrow L$ for Lower, U for Upper.

The matrix U is the upper triangular matrix obtained by applying Gaussian elimination (NOT Gauss-Jordan) to A . As long as no row swapping occurred when obtaining U , the decomposition $A = LU$ exists, and if $E_1, \dots, E_k$ were the elementary row operations applied, in that order, to A to produce U , then the matrix L can be obtained by applying the inverses of these operations in reverse order to the identity matrix I .

i.e.:

$$\begin{array}{ccc} A & \xrightarrow{E_1, \dots, E_k} & U \\ I & \xrightarrow{E_k^{-1}, \dots, E_1^{-1}} & L \end{array}$$

If E_i is the operation of adding row R_b to R_a , E_i^{-1} is the operation of subtracting row R_b from R_a .

If E_j is the operation of multiplying row R_k by μ , E_j^{-1} is the operation of multiplying row R_k by $1/\mu$.

If a row swap occurs, the LU decomposition does not exist $\rightarrow$ no row swapping for shifts 'n' gigs.

(Note - a PLU decomposition does exist for all square matrices, but we don't need to know how that works)

Example

Find the LU decomposition of $A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix}$

To find U :

$$\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix} \xrightarrow[\textcircled{2} R_3 - 7R_1]{\textcircled{1} R_2 - 4R_1} \begin{pmatrix} 1 & 2 & 3 \\ 0 & -3 & -6 \\ 0 & -6 & -12 \end{pmatrix} \xrightarrow{\textcircled{3} R_3 - 2R_2} \begin{pmatrix} 1 & 2 & 3 \\ 0 & -3 & -6 \\ 0 & 0 & 0 \end{pmatrix} \xrightarrow{\textcircled{4} -\frac{1}{3}R_2} \begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & -2 \\ 0 & 0 & 0 \end{pmatrix} = U$$

Gauss still requires leading 1s in the rows !!

To find L :

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \xrightarrow{-3R_2} \begin{pmatrix} 1 & 0 & 0 \\ 0 & -3 & 0 \\ 0 & 0 & 1 \end{pmatrix} \xrightarrow{R_3 + 2R_2} \begin{pmatrix} 1 & 0 & 0 \\ 0 & -3 & 0 \\ 0 & -6 & 1 \end{pmatrix} \xrightarrow[R_2 + 4R_1]{R_3 + 7R_1} \begin{pmatrix} 1 & 0 & 0 \\ 0 & -3 & 0 \\ 0 & -6 & 1 \end{pmatrix} = L$$

$$\therefore A = \begin{pmatrix} 1 & 0 & 0 \\ 4 & -3 & 0 \\ 7 & -6 & 1 \end{pmatrix} \begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & -2 \\ 0 & 0 & 0 \end{pmatrix}$$

LU Decomposition can be used in solving linear systems $Ax = b$:

If $A = LU$ exists:

$$LUx = b$$

by letting $Ux = y$:

$$Ly = b$$

Since L is triangular, the solutions y are easy to obtain. Substituting this y back into $Ux = y$ gives us a simple way of finding x . Since U is triangular.

Example

Solve $\begin{pmatrix} 2 & 6 & 2 \\ -3 & -8 & 0 \\ 4 & 9 & 2 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 2 \\ 2 \\ 3 \end{pmatrix}$ using LU decomposition

Finding U:

$$\begin{pmatrix} 2 & 6 & 2 \\ -3 & -8 & 0 \\ 4 & 9 & 2 \end{pmatrix} \xrightarrow{\frac{1}{2}R_1} \begin{pmatrix} 1 & 3 & 1 \\ -3 & -8 & 0 \\ 4 & 9 & 2 \end{pmatrix} \xrightarrow{\substack{R_2+3R_1 \\ R_3-4R_1}} \begin{pmatrix} 1 & 3 & 1 \\ 0 & 1 & 3 \\ 0 & -3 & -2 \end{pmatrix} \xrightarrow{R_3+3R_2} \begin{pmatrix} 1 & 3 & 1 \\ 0 & 1 & 3 \\ 0 & 0 & 7 \end{pmatrix} \xrightarrow{\frac{1}{7}R_3} \begin{pmatrix} 1 & 3 & 1 \\ 0 & 1 & 3 \\ 0 & 0 & 1 \end{pmatrix} = U$$

Finding L

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \xrightarrow{7R_3} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 7 \end{pmatrix} \xrightarrow{R_3-3R_2} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -3 & 7 \end{pmatrix} \xrightarrow{\substack{R_3+4R_1 \\ R_2-3R_1}} \begin{pmatrix} 1 & 0 & 0 \\ -3 & 1 & 0 \\ 4 & -3 & 7 \end{pmatrix} \xrightarrow{2R_1} \begin{pmatrix} 2 & 0 & 0 \\ -3 & 1 & 0 \\ 4 & -3 & 7 \end{pmatrix} = L$$

$$\begin{pmatrix} 2 & 0 & 0 \\ -3 & 1 & 0 \\ 4 & -3 & 7 \end{pmatrix} \begin{pmatrix} 1 & 3 & 1 \\ 0 & 1 & 3 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 2 \\ 2 \\ 3 \end{pmatrix}$$

$$\text{let } \begin{pmatrix} 1 & 3 & 1 \\ 0 & 1 & 3 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix} = \begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix}$$

$$\begin{pmatrix} 2 & 0 & 0 \\ -3 & 1 & 0 \\ 4 & -3 & 7 \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix} = \begin{pmatrix} 2 \\ 2 \\ 3 \end{pmatrix}$$

$$2y_1 = 2 \rightarrow y_1 = 1$$

$$-3y_1 + y_2 = 2 \rightarrow y_2 = 2 + 3 = 5$$

$$4y_1 - 3y_2 + 7y_3 = 3 \rightarrow 7y_3 = 3 - 4 + 15 \rightarrow y_3 = \frac{14}{7} = 2$$

Now solve for $\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}$:

$$\begin{pmatrix} 1 & 3 & 1 \\ 0 & 1 & 3 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 1 \\ 5 \\ 2 \end{pmatrix}$$

$$x_3 = 2$$

$$x_2 + 3x_3 = 5 \rightarrow x_2 = 5 - 6 = -1$$

$$x_1 + 3x_2 + x_3 = 1 \rightarrow x_1 = 1 + 3 - 2 = 2$$

$$\therefore x_1 = 2$$

$$x_2 = -1$$

$$x_3 = 2$$

Not too sure why anyone would do all this when Gauss-Jordan exists but apparently computers like it

LU decomposition can also be used to calculate the determinant of a matrix easily.

If $A = LU$, then $\det(A) = \det(L)\det(U)$, and since L and U are triangular, their determinants are the product of their diagonal entries.

$$\text{Using the previous example, } A = \begin{pmatrix} 2 & 6 & 2 \\ -3 & -8 & 0 \\ 4 & 9 & 2 \end{pmatrix} = \begin{pmatrix} 2 & 0 & 0 \\ -3 & 1 & 0 \\ 4 & -3 & 7 \end{pmatrix} \begin{pmatrix} 1 & 3 & 1 \\ 0 & 1 & 3 \\ 0 & 0 & 1 \end{pmatrix}$$

$$\det(A) = (2 \times 1 \times 7) \times (1 \times 1 \times 1) = 14$$